

The Fiber Laser Settings Bible



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L A S E R
S E T T I N G S

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Introduction and Purpose

The **Fiber Laser Settings Bible for LightBurn** is a complete, field-tested framework for preparing, optimizing, and applying fiber laser settings in LightBurn—starting with image preparation and ending with consistent, high-quality engravings across a wide range of materials.

Whether you're a hobbyist, workshop owner, or industrial engraver, this guide eliminates guesswork from your laser process. Built on hundreds of hours of calibration and testing, it helps you skip the frustrating phase of trial-and-error and jump straight to results that are repeatable, sharp, and professional.

This document focuses on one of the most important yet often misunderstood parts of laser engraving: preparing raster (bitmap) images. Raster images—such as photos, scanned logos, or textures—do not contain paths or vectors, which means they require special preparation before you can assign LightBurn settings such as 'Cut' or 'Fill'.

Here's what you'll learn:

- Why certain raster images fail when engraved without preparation.
- How to clean and enhance image contrast for laser clarity.
- How to convert your images to dithered black-and-white to preserve detail.
- How to trace and transform raster files into usable vectors within LightBurn.

Included in this workflow is a structured process to ensure every image you engrave:

- Comes out sharp and consistent.
- Responds predictably to the chosen material settings.
- Requires minimal adjustments before sending to the laser.

This guide is your foundation—once mastered, it can save you hours of time on every project, reduce wasted materials, and make your engravings stand out from the competition.

What the Package Contains

- **LightBurn Settings Files (.clb)**
 - A library of preset material profiles for various thicknesses and finishes (e.g., stainless steel, aluminum, composites).
 - **Installation & Usage Guide (PDF)**
 - Step-by-step instructions for importing configurations into LightBurn
 - Tips for initial calibration and focus verification
 - **Test File**
 - A standard test pattern to immediately verify that your settings are correct
 - Guidance on how to interpret your test results
 - **Maintenance**
 - Recommendations for lens care and laser power level checks
 - **Video Tutorial Access (Link)**
 - Visual demonstrations of LightBurn setup and fine-tuning for flawless results
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How These Settings Help You Achieve Consistent Results

1. **Guaranteed Reproducibility**
Tested across multiple material batches, these parameters deliver the same high quality day after day.
2. **Reduced Testing Time**
Eliminate hours of manual power-and-speed tuning—our package provides pre-optimized values.
3. **Uniform Engraving & Cutting Quality**
Perfect contrast, smooth edges, and consistent engraving depth across varied surfaces, thanks to the ideal balance of power (W) and speed (mm/s).
4. **Savings on Consumables & Energy**
Correct settings from the start minimize material waste and extend the life of your laser tube.
5. **Flexibility & Scalability**
Our modular library lets you easily add new materials or thicknesses without disrupting existing profiles.

Supported File Formats

1. **Vector (SVG, DXF, AI)**
 - **SVG:** An open standard ideal for complex elements, with support for groups and layers.
 - **DXF:** Autodesk's standard format, common in CAD programs—perfect for technical components.
 - **AI:** Adobe Illustrator's native format, preserving precise paths and separate cut/engrave layers.
 2. **Raster (JPEG, PNG, BMP)**
 - **JPEG:** Lossy compression suited for photographs but not recommended for fine line work.
 - **PNG:** Lossless compression with transparency support—useful for graphics with transparent backgrounds.
 - **BMP:** Uncompressed, large file sizes, rarely used in modern production workflows.
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Why We Recommend Vector Formats for Direct Assignment

- **Clearly Defined Paths**
Vectors consist of mathematically defined curves and lines, allowing LightBurn to apply “Cut” or “Fill” parameters directly to each object without ambiguity.
- **Faster Setup**
With SVG/DXF/AI files, speed and power settings can be attached directly to each layer or color—eliminating extra “Trace” steps required by raster images.
- **Precision and Quality**
Vector engraving or cutting yields smooth edges and sharp corners, free of pixelation or scaling artifacts.
- **Scalable Workflow**
Modifying line thickness or shape later is quick and preserves quality, making updates seamless.
- **LightBurn Compatibility**
LightBurn instantly recognizes vector elements as contours (for “Cut”) or closed regions (for “Fill”), ensuring error-free assignment of your optimized settings.

Raster Image Preparation Steps

To achieve high-quality engravings and markings with a fiber laser, raster images must be properly prepared before importing them into LightBurn. Follow these steps for clear, uniform results without unwanted artifacts.

1. Simplify Contrast and Background

- **Remove unnecessary details**
 - Open the image in an editor (Photoshop, GIMP, Affinity Photo).
 - Zoom in (200-300%) to inspect fine textures or digital noise that may affect engraving quality
 - Clean these areas with tools like "Clone" or "Heal" to avoid unwanted marks in the engraving.
- **Adjust contrast**
 - Increase the overall contrast until the difference between light and dark areas becomes clear (observe the histogram).
 - Use the "Levels" or "Curves" function to enhance only mid-tones without overexposing bright points.
- **Uniform the background**

- If the background is colored or textured, apply a "Fill" filter to make it uniformly light gray or pure white.
 - Check that main edges are distinct from the background; manually clean if necessary.
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2. Save in Black-and-White (Dithered) for Better Results

- **Convert to grayscale**
 - From the editor, convert the image to "Grayscale."
 - **Apply dithering**
 - Select a dithering algorithm (Floyd–Steinberg or Stucki) that converts smooth gradients into dot patterns.
 - Adjust the threshold level until essential elements (outlines, shadows) remain visible as dots.
 - **Export the file**
 - Save as PNG or BMP without additional compression.
 - Check file size: too large may complicate processing in LightBurn; too small may lose detail.
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3. Step-by-Step "Trace" in LightBurn

1. **Import the raster image**
 - In LightBurn, go to **File > Import** and select the prepared file (PNG/BMP dithered).
2. **Select the object**
 - Click on the imported image to activate it.
3. **Open the Trace window**
 - From the toolbar, click the **Trace Image** icon (or **Tools > Trace Image**).
4. **Set tracing parameters**
 - **Threshold:** Adjust the slider until the traced contour precisely matches desired edges.
 - **Cleanup:** Set a moderate level (2–5px) to remove unwanted small polygons.
 - **Detail:** Adjust to maintain thin lines (use values between 50–70%).
5. **Apply tracing**
 - Click **OK** and wait for processing. The vector result will appear over the image.
6. **Check and correct**

- Hide the original raster image from the Layers panel to inspect the vector.
 - Remove unwanted control points using **Node Edit**, and group segments as “Cut” or “Fill” as needed.
7. **Assign package settings**
 - In the **Cut Settings** window, select the traced layer and choose the suitable profile (engrave or mark) from your fiber laser settings library.
 8. **Save the project**
 - Save the LightBurn file (.lbrn) for future reference and quick repetition.

Importing and Assigning Settings

To make the most of your fiber laser settings package in LightBurn, follow these simple steps:

1. Where to Find the Library in LightBurn

1. Open LightBurn and go to **Library** → **Import...** in the top menu.
 2. In the dialog that appears, navigate to the .clb file provided in your package.
 3. Select it and click **Open** — you’ll see the imported materials appear in the **Library** panel on the right.
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2. How to Apply a Material Profile to a Vector Object

1. Draw or import your vector object (SVG, DXF, AI) onto the workspace.
 2. Click the object to select it.
 3. In the **Cuts/Layers** panel (bottom left), find the layer assigned to your object (identified by color or name, e.g., 'Layer 1').
 4. Double-click the layer’s colored circle to open the **Cut Settings** window.
 5. From the list of available profiles (imported in step 1), choose the appropriate material (e.g., “Stainless Steel 0.5 mm – Engrave”).
 6. Click **OK** to apply the profile to that layer.
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3. Verifying Parameters Before Engraving/Cutting

1. **Review Speed and Power** – Ensure the values shown in **Cut Settings** match the selected profile.
2. **Set Operation Order** – In the **Cuts / Layers** panel, drag layers up or down to determine execution order (usually engraving before cutting).
3. **Check Focus** – Use your laser’s focus tool to confirm the correct distance between the lens and the material.
4. **Preview the Job** – Press **Preview** (F12) to simulate the laser path and catch any overlaps or misconfigurations.
5. **Perform a Test Run** – For new materials, run a small test section to validate the results before launching the full job.

Common Error Messages & Quick Fixes

To keep your workflow smooth when using your fiber laser settings packages in LightBurn, here are some of the most frequent errors and how to resolve them immediately:

1. “Assign” Button Greyed Out

Cause: No vector object is selected, or the layer is inactive. LightBurn disables the “Assign” button when there’s no clear match between an object and Cut/Fill settings.

Quick Fix:

1. Make sure the object is a properly imported vector (SVG, DXF, AI), not a raster image.
 2. Click directly on the object to select it (its border will turn orange).
 3. In the **Cuts / Layers** panel, confirm the layer is colored and active. Once selected, the “Assign” button will become available.
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2. “Image Mode” vs. “Fill/Cut”

Cause: You imported a raster image and tried to apply an engrave or cut profile without tracing. LightBurn stays in “Image” mode when the file isn’t converted to vector.

Quick Fix:

1. Run **Trace Image** to convert the raster into vector (see your preparation and trace steps).

2. After tracing, hide or delete the original image, select the resulting vector, and then choose your profile in **Cut Settings** (e.g., “Engrave” or “Cut”).
 3. In the **Cuts / Layers** panel, ensure the layer type is correct—“Fill” for fills and “Line” (Cut) for outlines.
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3. USB Driver or COM Port Issues

Cause: LightBurn can’t detect the laser connection due to missing/outdated drivers or incorrect COM port settings.

Quick Fix:

1. **Check Device Manager** (Windows) or **System Information** (Mac) to see if the USB device appears and note its COM port.
2. **Reinstall the USB driver** for your laser (downloaded from the manufacturer’s website).
3. In LightBurn, go to **Edit** → **Device Settings**, select the correct COM port, and click **OK**. Then click **Connect** on the toolbar.
4. If one port (e.g., COM3) doesn’t work, try the next one (COM4), as new systems often reassign ports.

Tips for Quality Optimization

To get the highest-quality results from your fiber laser settings in LightBurn, keep the following best practices in mind:

1. Recommended Maximum DPI for Various Materials

- **Stainless Steel & Aluminum:** up to **500–600 DPI** for crisp markings; higher DPI risks overheating and blur.
 - **Anodized Metals:** **400–500 DPI** to balance contrast without degrading the anodized layer.
 - **Titanium & Alloys:** **300–400 DPI** to avoid excessive heat buildup and material discoloration.
 - **Plastics & Composites:** **300–350 DPI**—higher resolution can melt or warp edges.
 - **Ceramics & Glass:** **150–250 DPI**; these materials respond best to slower, low-heat passes.
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2. When and How to Use Air Assist

- **When to Use**
 - **Cutting:** Always—air assist blows away debris and prevents charring on the cut edges.
 - **Deep Engraving:** Recommended for depths >0.2 mm to clear smoke and maintain line definition.
 - **High-Speed Engraving:** Helps maintain focus by keeping the lens and work area clean.
 - **How to Set It Up**
 - Connect your air compressor or shop air line to the laser's air-assist port.
 - Set pressure to **0.1–0.3 MPa (15–45 PSI)**—insufficient pressure leaves debris, while excessive pressure may displace fine material.
 - Position the nozzle **5–10 mm** above the work surface, angled slightly toward the beam path.
 - Test on a scrap piece: look for clean, bright edges (for cutting) or uniform smoke evacuation (for engraving).
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3. Lens Recommendations for Fine Detail

- **50–60 mm Focal Length**
 - Ideal for **general engraving** and thin cuts (<1 mm).
 - Offers a **small spot size** (~0.08–0.1 mm), balancing depth of field and detail.
- **75–100 mm Focal Length**
 - Best for **ultra-fine engraving** (photo-quality marks) and micro-text.
 - Provides an even **smaller spot** (~0.05–0.07 mm) at the cost of a shallower focus zone.
- **150 mm and Above**
 - Use for **deep engraving** or **thicker cuts** where a larger spot and greater depth of field help maintain power over uneven surfaces.

Frequently Asked Questions (FAQ)

Q: “Can I engrave a JPEG directly?”

A: Not reliably. LightBurn treats JPEGs as raster images, which must be traced or converted before engraving to ensure consistent results. To prepare a JPEG:

1. Optimize contrast and convert to black-and-white (dithered).
2. Use **Trace Image** to generate vector outlines.
3. Assign your engrave or cut profile to the resulting vectors.

Untraced JPEGs often engrave poorly, producing inconsistent marks and confusing LightBurn’s fill/image modes.

Q: “How do I save my modified settings?”

A: After adjusting any parameter in **Cut Settings** (power, speed, passes, etc.), click **Save** at the bottom of that dialog to overwrite the current profile. To create a new custom profile:

1. In **Cut Settings**, make your changes.
2. Click the **plus (+)** icon next to the profile dropdown.
3. Give the new profile a descriptive name (e.g., “Steel 1.0 mm – High Contrast”).
4. Click **OK**.

Your custom profile will now appear in the **Library** under “User” and can be exported or shared as part of your `.clb` package.

Q: “What do I do if the library doesn’t show up in LightBurn?”

A: If your `.clb` file fails to appear after import:

1. **Confirm File Type** – Ensure you’re importing a `.clb` file, not `.lbrnlib` or another format.
2. **Re-import** – Go to **Library** → **Import...**, select the `.clb` file, then click **Open**.
3. **Restart LightBurn** – Close and reopen the application; imported libraries sometimes require a restart to refresh the panel.
4. **Check Library Panel** – In LightBurn’s workspace, expand the **Library** tab on the right; scroll through categories (e.g., “User”) to locate your profiles.
5. **Verify File Integrity** – If it still doesn’t load, re-download the `.clb` package or request a fresh copy—corrupt files won’t import.

Safety & Best Practices

Working with fiber lasers requires strict adherence to safety protocols and manufacturer guidelines to protect both operators and equipment. Follow these essential practices:

1. Personal Protective Equipment (PPE)

- **Laser Safety Goggles:** **Always** wear laser safety goggles specifically rated for your laser's wavelength and power—standard safety glasses **do not** provide sufficient protection.
 - **Protective Clothing:** Wear non-reflective, fire-resistant garments and closed-toe shoes. Avoid jewelry or loose fabrics that could catch stray reflections.
 - **Gloves:** Use heat-resistant gloves when handling freshly engraved or cut parts, as they can become hot.
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2. Proper Ventilation & Smoke Management

- **Exhaust System:** Install a dedicated fume extractor or ducted ventilation to remove smoke, particulates, and vapors at the source.
 - **Airflow Rate:** Ensure airflow is strong enough to clear smoke without stirring it back onto the workpiece—typically 200–300 CFM for small enclosures.
 - **Filter Maintenance:** Replace or clean filters regularly according to the manufacturer's schedule to maintain extraction efficiency.
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3. Follow Manufacturer Recommendations

- **Operating Parameters:** Use only the approved power, speed, and duty-cycle settings specified for your laser model to prevent overheating or damage.
- **Maintenance Schedule:** Adhere to routine maintenance tasks—lens cleaning, alignment checks, and firmware updates—as outlined in your laser's user manual.
- **Warranty Compliance:** Deviating from recommended settings or modifying hardware may void warranties. Always consult the manufacturer before making changes.